FORECASTING / PRODUCTION MLOPS

From a forecasting notebook to repeatable planning inputs

Make store-level forecasts available through an API, monitor quality and release updates with checks and a recovery path.

WHY MOVE BEYOND THE NOTEBOOK?

01

Reliable delivery

Make forecasts available to planning applications without a manual notebook run and file export.

02

Less manual work

Automate repeatable data checks, training and release steps, with traceable results.

03

Controlled change

Monitor quality, evaluate new candidates and recover when release verification fails.

PROOF FROM A WORKING REFERENCE SYSTEM

65.8% lower RMSE

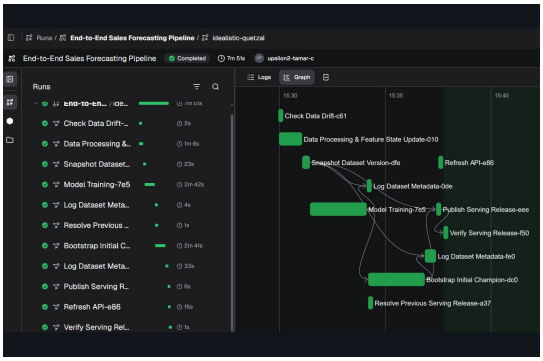
Affected promo stores, post-promotion comparison with a static baseline.

1 of 3 approved

Candidate training is separate from the decision to promote a model.

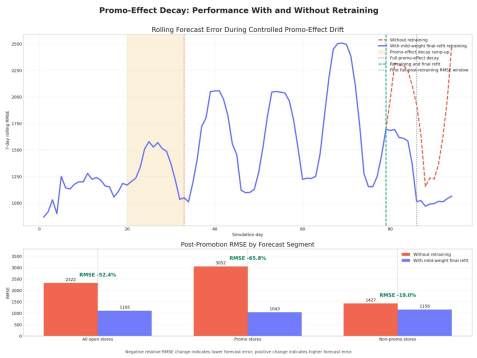
Verified recovery

Serving-release checks and a tested Cloud Run revision rollback.



Checked model delivery

Executed training, evaluation and serving-release verification workflow.



Measured adaptation

Controlled promotion-effect decay experiment and segment-level comparison.

WHAT WAS BUILT

FastAPI forecasting service, Prefect pipelines, MLflow registry, versioned serving bundles, model monitoring and Grafana service metrics. Container deployment to Cloud Run with CI/CD and Terraform.

EVIDENCE SCOPE

Independent reference project using public Rossmann data. The 65.8% result applies to affected promo stores in the post-promotion simulation window, not to a client deployment. Cloud deployment and rollback successfully verified.

Have a working model, but no reliable deployment?

Start with a Notebook-to-MLOps Assessment: a code review and prioritized delivery roadmap.

Steffen Lauterbach | Freelance MLOps Engineer

GitHub: mlops-sales-forecasting

LinkedIn / Contact